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SARDAR PATEL UNIVERSITY**Master of Science (Information Technology) (Semester 3) Examination 2025****PS03EINT56 : Software Testing****Date: 12/04/2025 (Saturday)****Time: 2:00 pm to 5:00 pm****Total: 70 Marks**

NOTES: Figure to the right indicate full marks of the questions.

Q-1 Choose the most appropriate option for each question:**[8]**

- (i) Software testing is an important because -
 - [A] it helps in saving money
 - [B] it ensures quality of a software product
 - [C] it offers customer satisfaction
 - [D] all of these
- (ii) Which one of the following terms is not related to software testing?
 - [A] test case [B] Error [C] Fault [D] test bot
- (iii) The Testing of the software with real data in real environment is known as _____ testing.
 - [A] Alpha [B] Beta [C] Dummy [D] System
- (iv) Boundary value analysis testing technique is an example of _____ testing.
 - [A] White-box [B] Black-box
 - [C] Both [A] and [B] [D] None of these
- (v) Software testing activities should be started:
 - [A] After the completion of source code
 - [B] After the completion of design phase
 - [C] As early as possible in the software development life cycle
 - [D] After the completion of software requirements and analysis phase
- (vi) Acceptance testing is done by -
 - [A] Testers [B] Programmers [C] Customers [D] Developers
- (vi) Which of the followings has a unique identification number?
 - [A] Test [B] Test Case [C] Test Suite [D] Test Data
- (viii) Which programming languages are supported by Selenium?
 - [A] Ruby [B] Java [C] Python [D] All of these

Q-2 Attempt the following: (ANY SEVEN)**(14)**

- i. Write down the benefits of Software Testing.
- ii. Define the terms: 'Fault' and 'Failure'.
- iii. Differentiate: Verification and Validation.
- iv. Differentiate: Black-Box Testing and White-Box Testing.

- v. What is control flow testing? Why is it used?
- vi. Justify – 'Software Testing is critical and time consuming activity'.
- vii. List down parameters on which classification of testing tools can be done?
- viii. Explain briefly on types of testing tools.
- ix. Give at least four names of software testing tools.

- Q-3 (a)** Define the term 'Testing'. Explain various levels of testing. (6)
- (b)** Explain – 'Testing Process' (6)

OR

- (b)** What is STLC? List and explain various phases/stages of STLC. (6)

- Q-4 (a)** What is 'Black-Box Testing'? How to do black-box testing? Explain any one black-box testing technique with an example. (6)
- (b)** Explain – 'Statement Coverage' and 'Path Coverage' testing techniques. (6)

OR

- (b)** Define the term 'White-Box Testing'. How do you perform white-box testing? Write down advantages and disadvantages of white-box testing. (6)

- Q-5 (a)** What is Test Case? Explain different test case parameters with an example. (6)
- (b)** Explain – 'Automated Test Data Generation'. (6)

OR

- (b)** Explain – 'Test Case Design Techniques'. (6)

- Q-6 (a)** What you mean by 'Manual Testing' and 'Automated Testing'? Write down the role of testing tools in software testing process. (6)
- (b)** What is Selenium? Write down various features and components of any software testing tool. (6)

OR

- (b)** Write down advantages and disadvantages of testing tools. (6)

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